TITLE:FRAUD DETECTION USING MACHINE LEARNING

SOURCE CODE:

```
Fraud-Detection-ML/
│
├── dataset/
│   └── creditcard.csv
│
├── model/
│   └── fraud_model.pkl
│
├── train_model.py
├── predict.py
├── app.py
├── templates/
│   └── index.html
├── static/
│   └── style.css
├── requirements.txt
└── README.md
```

1.train_model.py

```
import pandas as pd
import pickle

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score

# Load dataset
data = pd.read_csv("dataset/creditcard.csv")

# Features and Target
X = data.drop("Class", axis=1)
y = data["Class"]

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
   X,
```

```
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

# Train Model
model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)

model.fit(X_train, y_train)

# Test Accuracy
prediction = model.predict(X_test)

accuracy = accuracy_score(y_test, prediction)

print("Model Accuracy :", accuracy)

# Save model
pickle.dump(model, open("model/fraud_model.pkl", "wb"))

print("Model Saved Successfully")
```

2. predict.py

```
import pickle
import numpy as np

model = pickle.load(open("model/fraud_model.pkl", "rb"))

def predict_transaction(features):

    features = np.array(features).reshape(1, -1)

    result = model.predict(features)

    if result[0] == 1:
        return "Fraud Transaction"
```

```
    else:
        return "Legitimate Transaction"
```

3. app.py

```
from flask import Flask, render_template, request
import pickle
import numpy as np

app = Flask(__name__)

model = pickle.load(open("model/fraud_model.pkl", "rb"))

@app.route("/")
def home():
    return render_template("index.html")


@app.route("/predict", methods=["POST"])

def predict():

    values = []

    for i in range(30):
        values.append(float(request.form[f"f{i}"]))

    final = np.array(values).reshape(1, -1)

    prediction = model.predict(final)

    if prediction[0] == 1:
        output = "Fraudulent Transaction"

    else:
        output = "Legitimate Transaction"

    return render_template("index.html", prediction_text=output)


if __name__ == "__main__":
    app.run(debug=True)
```

4. templates/index.html

```
<!DOCTYPE html>

<html>

<head>

<title>Fraud Detection</title>

<link rel="stylesheet" href="{{ url_for('static', filename='style.css') }}">

</head>

<body>

<div class="container">

<h1>Credit Card Fraud Detection</h1>

<form action="/predict" method="post">

{% for i in range(30) %}

<input type="text"
name="f{{i}}"
placeholder="Feature {{i}}"
required>

{% endfor %}

<br>

<button type="submit">Predict</button>

</form>

<h2>

{{ prediction_text }}
```

```
</h2>

</div>

</body>

</html>
```

5. static/style.css

```
body{

background:#f0f2f5;

font-family:Arial;

}

.container{

width:700px;

margin:auto;

background:white;

padding:30px;

margin-top:40px;

border-radius:10px;

box-shadow:0px 0px 10px gray;

}

h1{

text-align:center;

}
```

```
input{
width:95%;
padding:10px;
margin:5px;
}
button{
padding:12px;
width:100%;
background:#0b7dda;
color:white;
font-size:18px;
border:none;
cursor:pointer;
}
button:hover{
background:#005fa3;
}
h2{
text-align:center;
color:green;
}
```

6. requirements.txt

Flask
pandas
numpy
scikit-learn
pickle-mixin

Install using:

pip install -r requirements.txt

7. README.md

Credit Card Fraud Detection Using Machine Learning

Description

This project detects fraudulent credit card transactions using a Random Forest Classifier.

Dataset

Kaggle Credit Card Fraud Detection Dataset

creditcard.csv

Technologies

Python

Flask

Pandas

NumPy

Scikit-learn

HTML

CSS

Project Workflow

Load Dataset

↓

Preprocess Data

↓

Train Random Forest Model

↓

Save Model

↓

Load Model

↓

Predict Transaction

↓

Display Result

Run Project

Train Model

python train_model.py

Run Flask App

python app.py

Open Browser:

http://127.0.0.1:5000

How to Run

1. Download the **`creditcard.csv`** dataset from Kaggle and place it in the `dataset/` folder.
2. Install the required packages:

   pip install -r requirements.txt

3. Train the model:

   python train_model.py

   This creates `model/fraud_model.pkl`.

4. Start the Flask application:

   python app.py

5. Open your browser and go to:

   http://127.0.0.1:5000